

BEYOND SPEED RANKINGS: A NINE-COUNTRY MEASUREMENT STUDY OF SOUTHEAST ASIA’S APPLICATION-LAYER DIGITAL DIVIDE

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990M

Speed tests analyzed

9

SE Asian countries

100%

HD-video success, every country

<38%

Cloud-gaming success, weakest markets

Motivation

Speed-index rankings say little about whether real applications work. Thailand ranked **12th worldwide** for fixed broadband (Ookla Speedtest Global Index, June 2026), yet a 2023 Thailand Consumer Council survey found **81%** of 2,924 respondents had connection problems — mostly slow and dropping connections. The mismatch is not unique to Thailand.

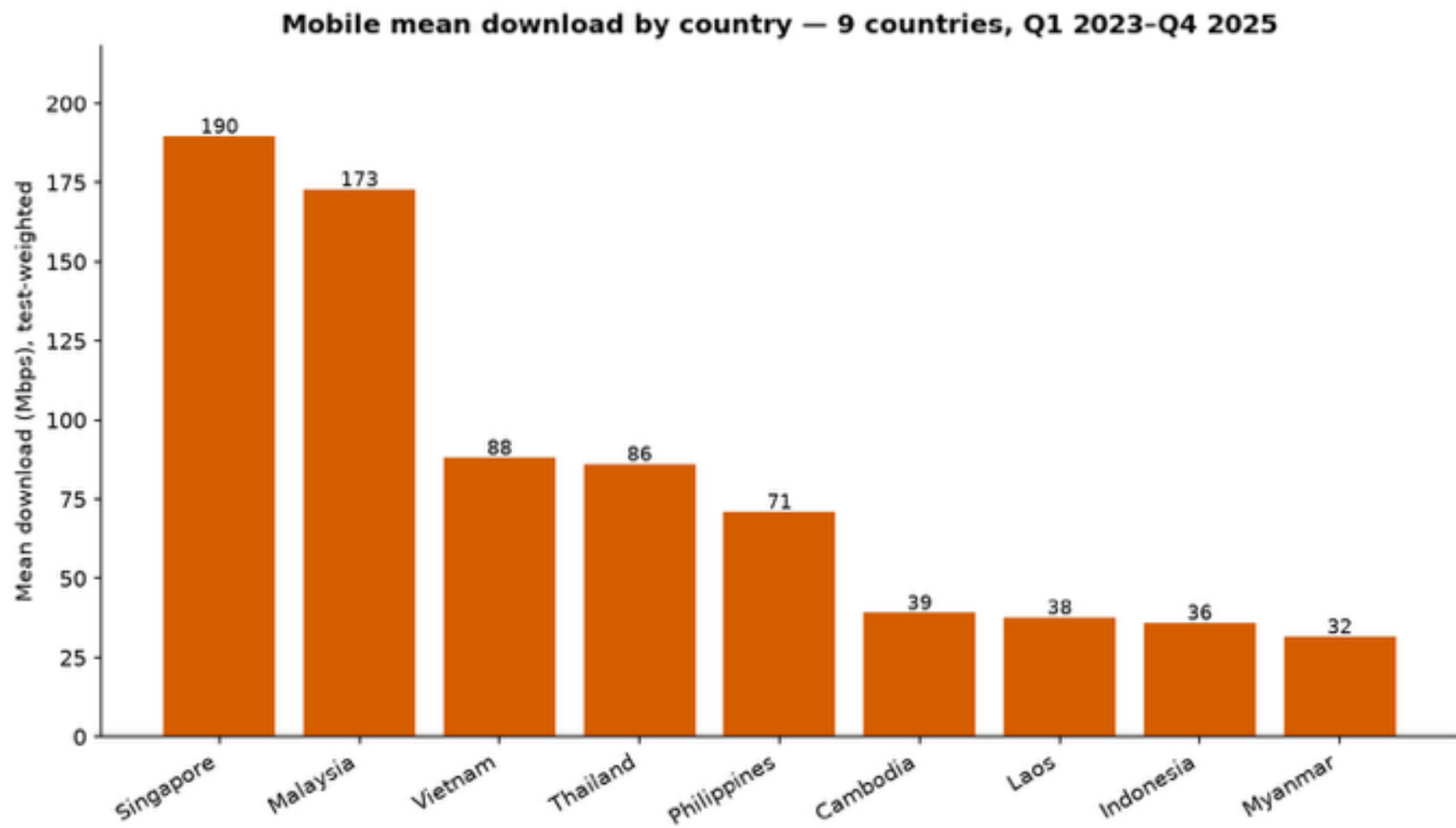
Question: how do nine Southeast Asian networks perform against the demands of real applications — voice, HD video, UHD video and cloud gaming?

Data

- **Ookla Open Data:** 229.4M tests, quarterly ~600 m tiles, fixed + mobile
- **M-Lab NDT7:** 760.6M single-stream tests
- **Coverage:** Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, Philippines, Malaysia, Singapore; Q1 2023–Q4 2025
- Tiles mapped to provinces (point-in-polygon, geoBoundaries ADM1), test-weighted; province-quarters with ≥ 100 tests and ≥ 5 tiles
- Operators grouped by **ASN**, not raw ISP names

Mobile Is Uniform, Fixed Is Not

Fixed broadband beats mobile in most countries: Thailand shows the largest gap (**3.2×**, ~275 vs. ~85 Mbps), Singapore 2.1×. Myanmar is the only country where mobile is *faster* than fixed (0.8×) — a mobile-first market.



Mobile mean download by country, Q1 2023–Q4 2025.

Leading mobile operators in seven of nine markets sit in a narrow **22–39 Mbps** band, versus fixed operators spanning 23–300 Mbps.

Data & Code



github.com/chissanupun/
thailand-broadband-analysis

Finding 1 — Basics Are Solved

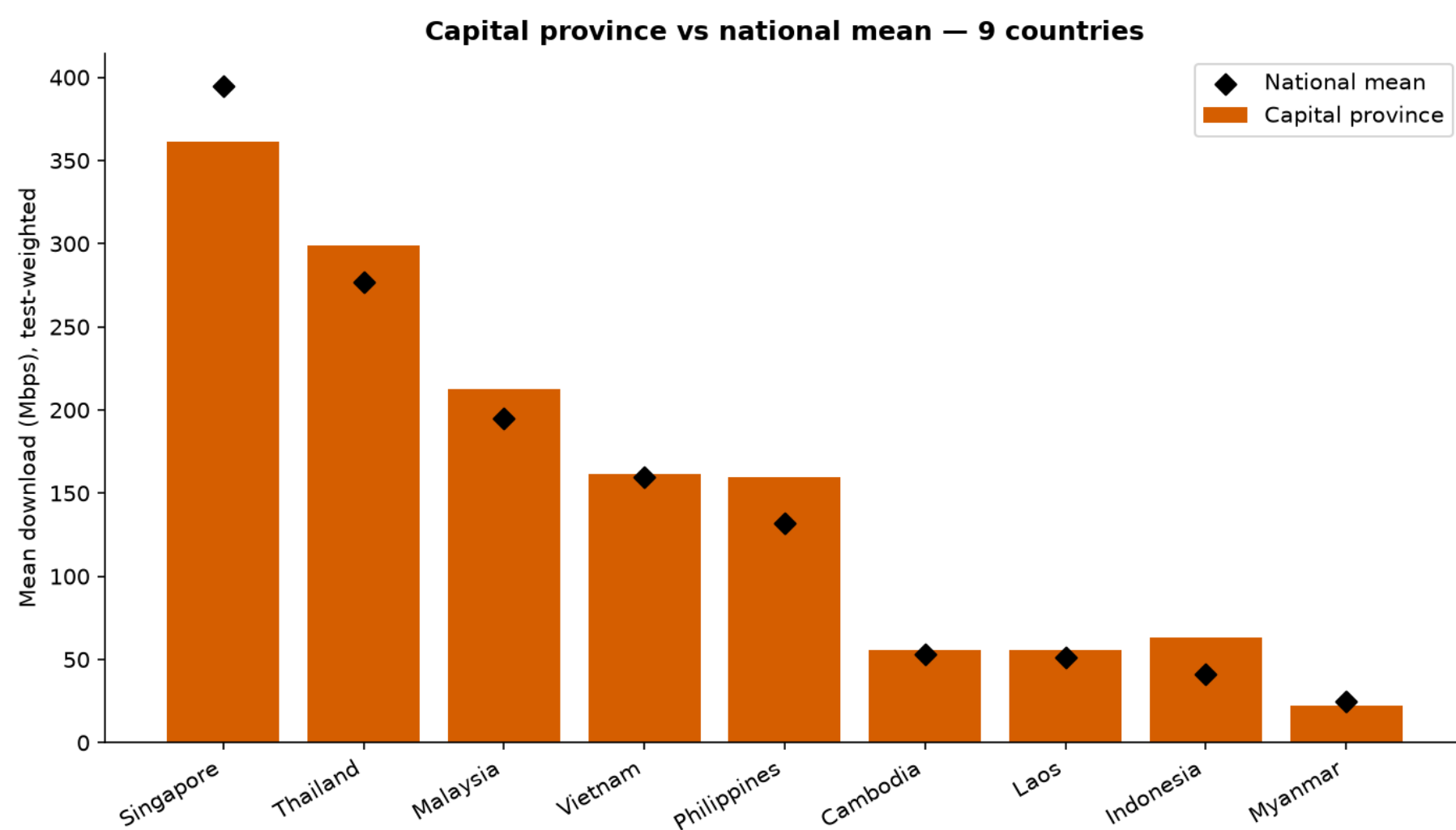
Voice (≥ 64 kbps) and HD video (≥ 5 Mbps) succeed at **100%** in every country, every quarter, on fixed and mobile networks. Even the lowest-tier broadband markets meet the basic floor.

Application	Data rate	Latency
Voice	64 kbps	200 ms
Video (HD)	5 Mbps	few s
Video (UHD)	25 Mbps	few s
Cloud gaming	44 Mbps	25 ms

Thresholds after Lübben & Misfeld (2022). Success rate = national-level share meeting the download-speed threshold, per quarter.

Finding 2 — Urban–Rural Stratification

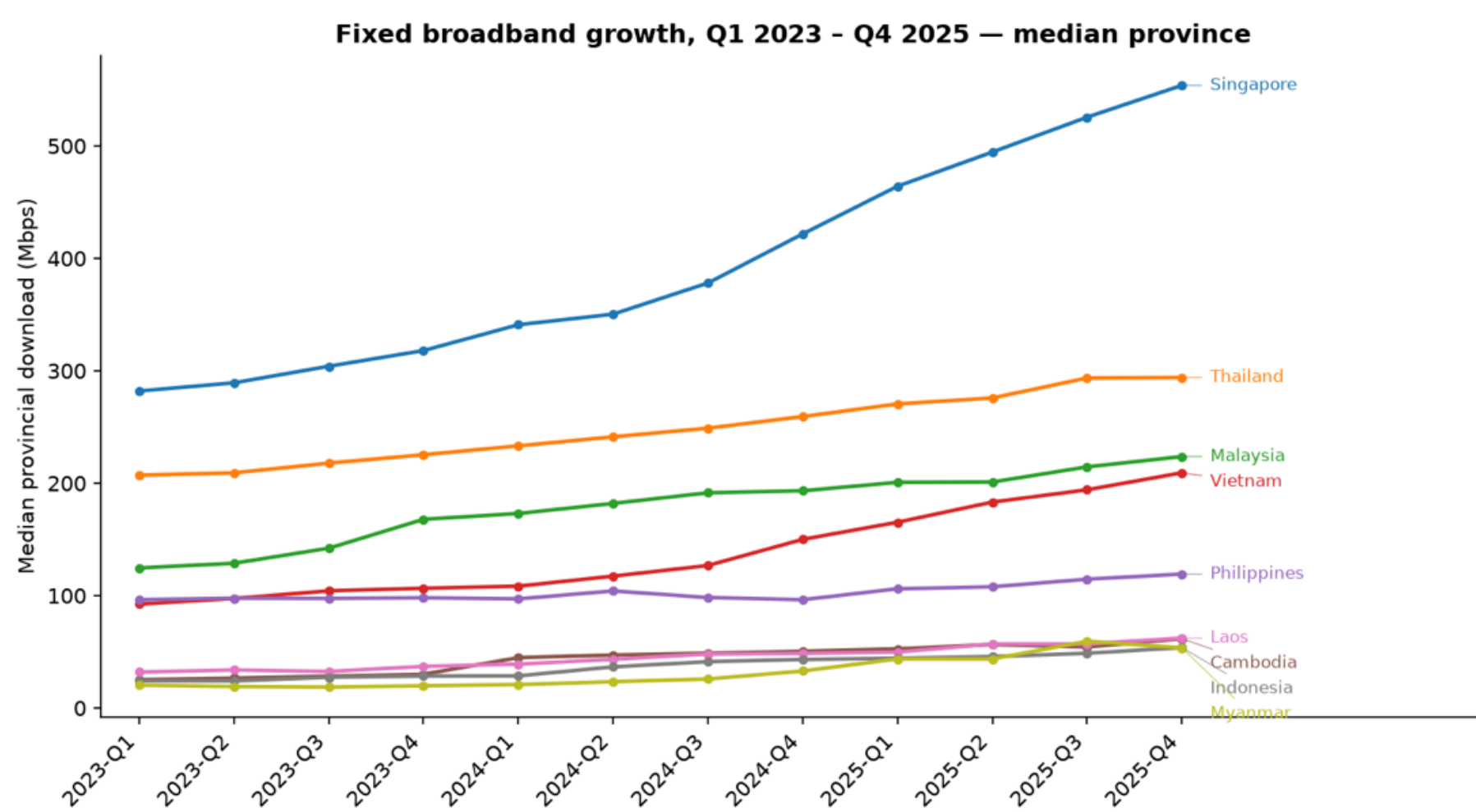
Most capitals outperform their nation: Bangkok 298.8 vs. 276.9 Mbps, Kuala Lumpur 212.3 vs. 194.8, Manila NCR 159.6 vs. 132.0. Singapore’s Central Region (361.0) sits *below* its national mean (394.8). Peak capital speed is **over 16×** the lowest capital (Myanmar, 22.0 Mbps).



Capital province vs. national mean, fixed broadband, test-weighted.

Finding 3 — Speed Gap Is Widening

Singapore nearly **doubled** its median provincial speed (~280 → ~550 Mbps); Thailand grew ~200 → ~300. Vietnam accelerated to catch Malaysia (200+ Mbps). Laos, Cambodia, Indonesia and Myanmar stayed **below 60 Mbps** for all three years.

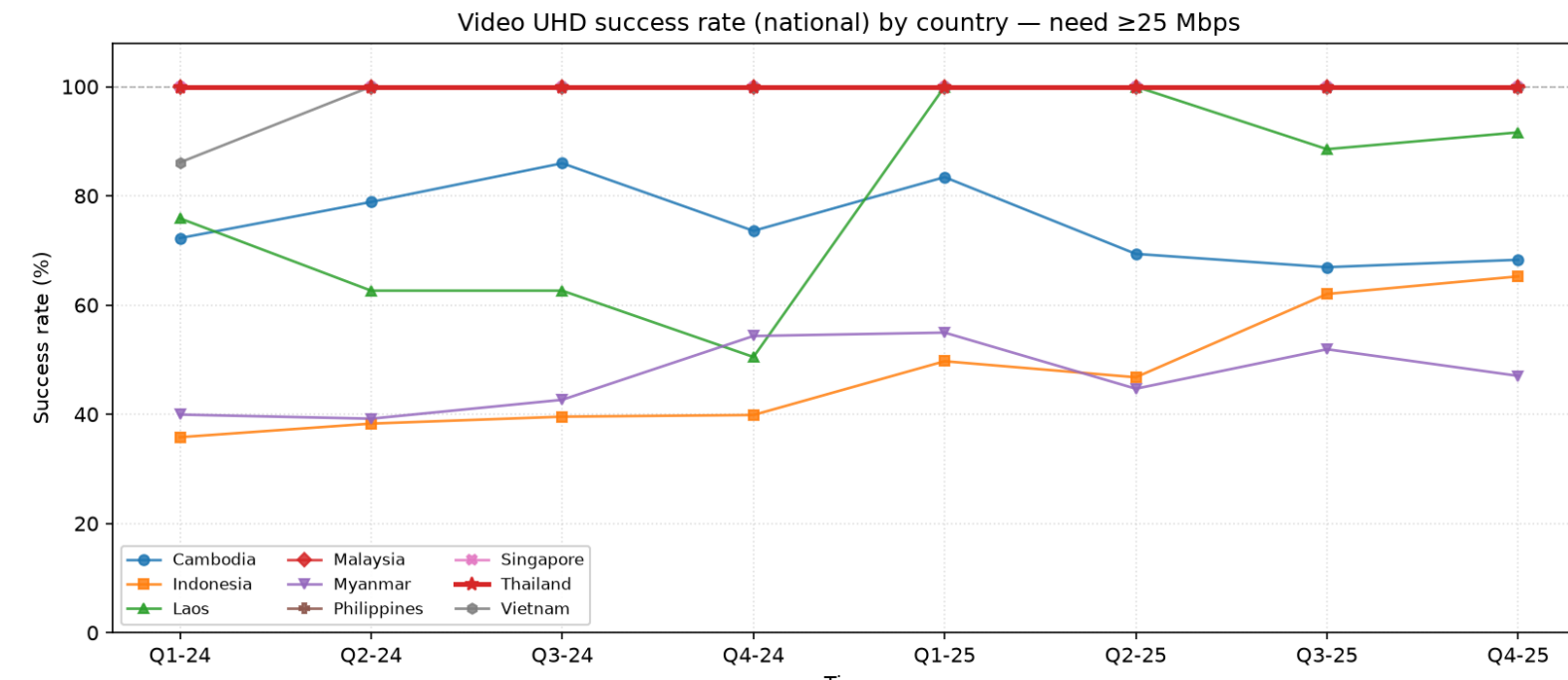


Median provincial fixed download speed, Q1 2023–Q4 2025.

Finding 4 — Application Tiers Diverge

UHD video (≥ 25 Mbps)

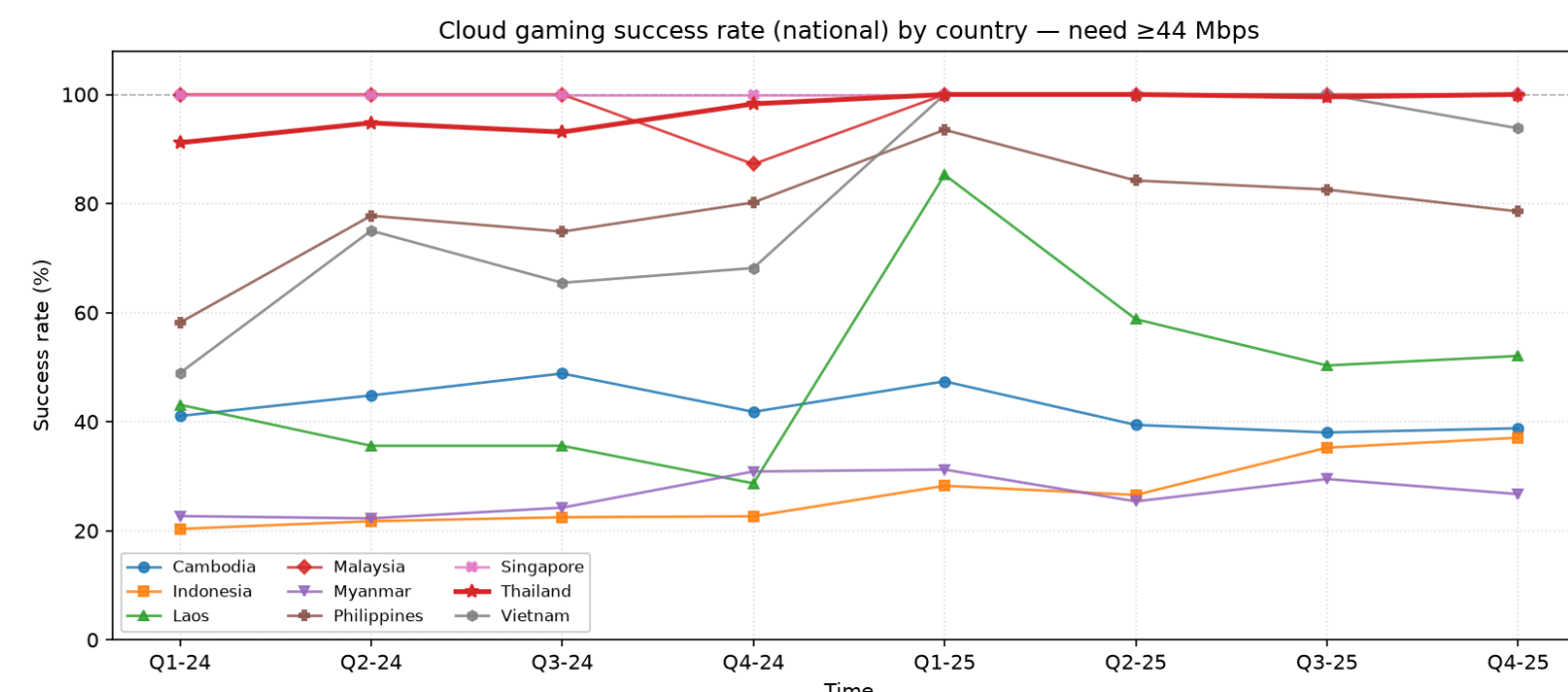
- Malaysia, Singapore, Thailand, Philippines, Vietnam: 100% every quarter
- Cambodia 67–86%; Indonesia 36 → 65%; Myanmar 40–55%
- Laos volatile: 50–100%



UHD-video success rate by country.

Cloud gaming (≥ 44 Mbps)

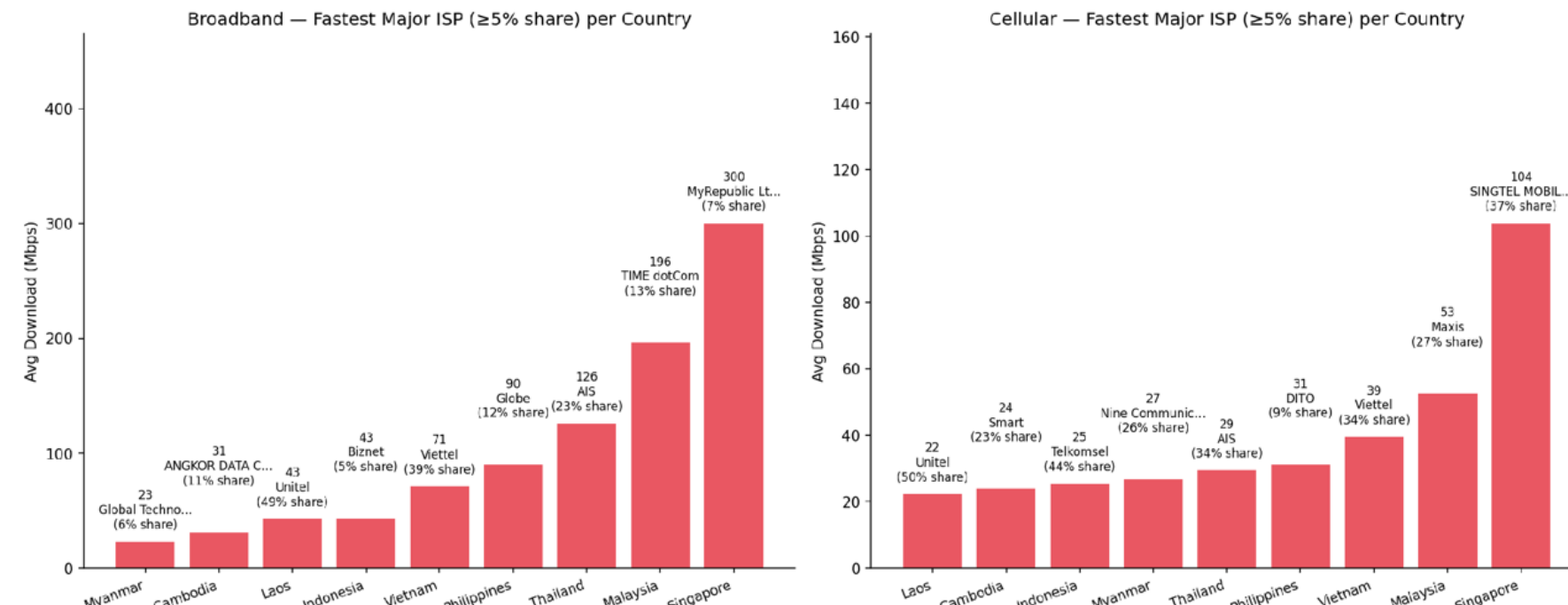
- Singapore ~100%; Thailand and Malaysia 87–100%
- Philippines 58–93%; Vietnam <50% → ~94%
- Cambodia 38–49%; Laos ~52%
- Indonesia <38%; Myanmar <31%



Cloud-gaming success rate by country.

Finding 5 — Who Delivers the Speed

Fixed: **MyRepublic** (SG) 300 Mbps, TIME dotCom (MY) 196, AIS (TH) 126; everyone else below 100, down to 23 (Myanmar). The *most-used* fixed ISP ranges from Singtel Fibre **176** Mbps to Telkom Indonesia **19** and Mytel Myanmar **10**.



Fastest major ISP ($\geq 5\%$ share) per country, fixed (left) and cellular (right).

Takeaway

Raw speed rankings hide the divide that matters: not whether a country “has broadband,” but whether it survives a real application. Basic needs are met everywhere; UHD and cloud gaming expose the gap. This argues for targeted fiber deployment, regional investment and transport-layer optimization.